Física 3 – Semana 04: Calor sensível – Resolução

Prof. Vogt

1, B

2. C

$$d = \frac{m}{V}$$

$$1{,}2 = \frac{m}{1{,}5 \cdot 10^{-3}}$$

$$m = 1{,}8 \cdot 10^{-3}\ kg$$

$$\boxed{m = 1{,}8\ g}$$

$$Q = m \cdot c \cdot \Delta\theta$$

$$P = \frac{E}{\Delta t}$$

$$P = \frac{1{,}8 \cdot 0{,}24 \cdot (30 - 5)}{1}$$

$$P = 10{,}8\ cal/s \qquad \times 4{,}2$$

$$\boxed{P = 45{,}36\ W}$$

3.

a) $d = \frac{m}{V}$

$1{,}26 = \frac{m}{200}$

$\boxed{m_g = 252\ g}$

$Q = m \cdot c \cdot \Delta\theta$

$Q = 252 \cdot 0{,}6 \cdot (70 - 20)$

$\boxed{Q = 7560\ cal}$

b) $\Delta V = V_0 \cdot \gamma \cdot \Delta\theta$

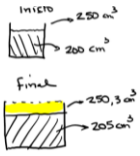
$\Delta V_g = 200 \cdot 5 \cdot 10^{-4} \cdot (70 - 20)$

$\boxed{\Delta V_g = 5\ cm^3}$

$V_{vazia} = V_{frasco} - V_{líq}$

$V_{vazia} = 250{,}3 - 205$

$\boxed{V_{vazia} = 45{,}3\ cm^3}$



4. B

$$\underbrace{Q_1}_{m_1 \cdot c_1 \cdot \Delta\theta_1} + \underbrace{Q_2}_{m_2 \cdot c_2 \cdot \Delta\theta_2} = 0$$

$$m_1 \cdot 4200 \cdot (70 - 80) + 0{,}42 \cdot 900 \cdot (70 - 20) = 0$$

$$-42000 \cdot m_1 + 42 \cdot 9 \cdot 50 = 0$$

$$\cancel{42}000 \cdot m_1 = \cancel{42} \cdot 450$$

$$m_1 = 0{,}45\ kg$$

$$\boxed{m_1 = 450\ g}$$

5. C

$$P = \frac{E}{\Delta t} \qquad Q = m \cdot c \cdot \Delta\theta$$

$$P = \frac{300 \cdot 1 \cdot (40 - 20)}{2}$$

$$\boxed{P = 3000\ cal/min}$$

$$P = \frac{E}{\Delta t}$$

$$\cancel{3000} = \frac{\cancel{300} \cdot 1 \cdot (80 - 40) + \cancel{300} \cdot 1 \cdot (80 - 20)}{\Delta t}$$

$$10 \cdot \Delta t = 60 + 40$$

$$\boxed{\Delta t = 10\ min}$$

6. C

$$\underbrace{Q_1}_{m_1 \cdot c_1 \cdot \Delta\theta_1} + \underbrace{Q_2}_{C_2 \cdot \Delta\theta_2} + \underbrace{Q_3}_{m_3 \cdot c_3 \cdot \Delta\theta_3} = 0$$

$$200 \cdot 0{,}03 \cdot (25 - \theta) + 46 \cdot (25 - 20) + 200 \cdot 1 \cdot (25 - 20) = 0$$

$$6 \cdot (25 - \theta) + 46 \cdot 5 + 200 \cdot 5 = 0$$

$$150 - 6 \cdot \theta + 230 + 1000 = 0$$

$$6 \cdot \theta = 1380$$

$$\boxed{\theta = 230°C}$$

7. C

$$Q_1 + Q_2 = 0$$

$$m_1 \cdot c_1 \cdot \Delta\theta_1 + m_2 \cdot c_2 \cdot \Delta\theta_2 = 0$$

$$18000 \cdot 1 \cdot (40 - 20) + 12000 \cdot 1 \cdot (T - 85) = 0$$

$$6000 \cdot (60 + 2T - 170) = 0$$

$$2T = 110$$

$$\boxed{T = 55°C}$$

8. B

$$P = \frac{E}{\Delta t}$$

$$P = \frac{m \cdot c \cdot \Delta\theta}{\Delta t}$$

$$\boxed{\Delta t = \frac{m \cdot c \cdot \Delta\theta}{P}}$$

$$P' = \frac{E'}{\Delta t'}$$

$$P + \frac{P}{3} = \frac{m \cdot c \cdot \Delta\theta}{\Delta t'}$$

$$\frac{4 \cdot P}{3} = \frac{m \cdot c \cdot \Delta\theta}{\Delta t'}$$

$$\Delta t' = \frac{3 \cdot m \cdot c \cdot \Delta\theta}{4 \cdot P} \qquad \Delta t$$

$$\boxed{\Delta t' = \frac{3}{4}\Delta t}$$

9. E

Cálculo da potência útil:

$$P = 2 \cdot 10^3\ \frac{W}{m^2} \cdot 20\ m^2 \cdot 0{,}6 = 24000\ W$$

A quantidade de calor trocada é dada por:

$$Q = m \cdot c \cdot \Delta\theta = m \cdot 1 \cdot 4{,}2 \cdot \Delta\theta$$

Substituindo esse resultado na equação abaixo, vem:

$$P = \frac{Q}{\Delta t} = \frac{m}{\Delta t} \cdot 4{,}2 \cdot \Delta\theta$$

Como:

$$\frac{m}{\Delta t} = 6\ \frac{\ell}{min} = \frac{6}{60}\ \frac{kg}{s}$$

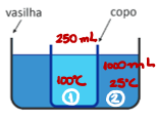
Temos que:

$$24000 = \frac{6}{60} \cdot 4{,}2 \cdot \Delta\theta$$

$$\therefore \Delta\theta \cong 57{,}1\ °C$$

10. B

Passo 1:



$$Q_1 + Q_2 = 0$$

$$m_1 \cdot c_1 \cdot \Delta\theta_1 + m_2 \cdot c_2 \cdot \Delta\theta_2 = 0$$

$$d_A \cdot 250 \cdot 1 \cdot (\theta_E - 100) + d_A \cdot 1000 \cdot 1 \cdot (\theta_E - 25) = 0$$

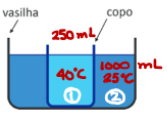
$$d_A \cdot 250 \cdot [\theta_E - 100 + 4\theta_E - 100] = 0$$

$$5 \cdot \theta_E - 200 = 0$$

$$5 \cdot \theta_E = 200$$

$$\boxed{\theta_E = 40°C}$$

Passo 3:



$$Q_1' + Q_2' = 0$$

$$m_1 \cdot c_1 \cdot \Delta\theta_1' + m_2 \cdot c_2 \cdot \Delta\theta_2' = 0$$

$$d_A \cdot 250 \cdot 1 \cdot (\theta_E' - 40) + d_A \cdot 1000 \cdot 1 \cdot (\theta_E' - 25) = 0$$

$$d_A \cdot 250 [\theta_E' - 40 + 4\theta_E' - 100] = 0$$

$$5\theta_E' - 140 = 0$$

$$5\theta_E' = 140$$

$$\boxed{\theta_E' = 28°C}$$

11. B

$$P = \frac{E}{\Delta t} \qquad Q = m \cdot c \cdot \Delta\theta$$

$$P_{útil} = \frac{500 \cdot 4{,}2 \cdot (100 - 20)}{3 \cdot 60}$$

$$\boxed{P_{útil} \approx 933\ W}$$

$$\eta = \frac{P_{útil}}{P_{total}}$$

$$\eta = \frac{933}{1200}$$

$$\boxed{\eta \approx 75\ \%}$$

12. D

$$Q = mc\Delta T$$

$$Q = 700 \cdot 1 \cdot 15$$

$$Q = 10500 \text{ cal}$$

$$Q = 10500 \times 4,2$$

$$Q = 44100\ J$$

$$P_{ref} = \frac{Q}{t}$$

$$P_{ref} = \frac{44100}{420}$$

$$P_{ref} = 105\ W$$

$$0,21P = 105$$

$$P = \frac{105}{0,21}$$

$$P = 500\ W$$